R15

Code No: 121AL

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year Examinations, September/October - 2021
MATHEMATICAL METHODS
(Common to EEE, ECE, CSE, EIE, IT)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
- - -

JNTUH USED SEPT-2021

1.a) Find the second difference of the polynomial $x^4 - 12x^3 + 42x^2 - 30x + 9$ with interval of difference h=2.

b) Fit a curve y = a + bx by the method of least squares.

x	1	2	3	4	5	6	7	8	9
y	9	8	10	12	11	13	14	16	15

[7+8]

2.a) Find y(3.4) from the following table using Newton's forward interpolation formula.

x	3	4	5	6
y	31	69	131	223

b) Fit a curve using Lagrange's interpolation formula.

x	0	5	10	15	20	25
y	12	15	17	22	24	30

[7+8]

3.a) Find a root of the equation $f(x) = x + \log x - 2$ using Newton Raphson method.

b) Given that $\frac{dy}{dx} = 1 + xy$ and y(0)=1. Compute y(0.1) using Picards method upto three decimals. [7+8]

4. Find y(0.1) and y(0.2) using Runge Kutta method given that $y' = x^2 - y, \quad y(0) = 1$. [15]

5. Find the Fourier sine transform of $\frac{1}{x\left(a^2 + x^2\right)}$. [15]

6. Obtain a Fourier series for $f(x) = x \sin x$ in $(0 \le x \le 2\pi)$. [15]

7. Solve the following partial differential equations:
   a) $z^2(p^2 + q^2 + 1) = c^2$
   b) $x(y-z)\ p + y(z-x)q = z(x-y)$. [8+7]

8. Verify Green's theorem for $\int_c (3x - 8y^2)dx + (4y - 6xy)dy$ where C is the region bounded by $x = 0,\ y = 0\ and\ x + y = 1$. [15]

---ooOoo---